

sdstudio

A Companion Package for Designing
and Managing surveydown Surveys

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About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
 1. **Data Science** in sustainable transportation and human behavior
 2. **Open-Source Software** in  tools for survey research
 3. **Agentic AI Engineering** around Claude Code
- For more information, visit pingfanhu.com

jhelvy.shinyapps.io



Welcome to the Ultimate Superhero Survey!



Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

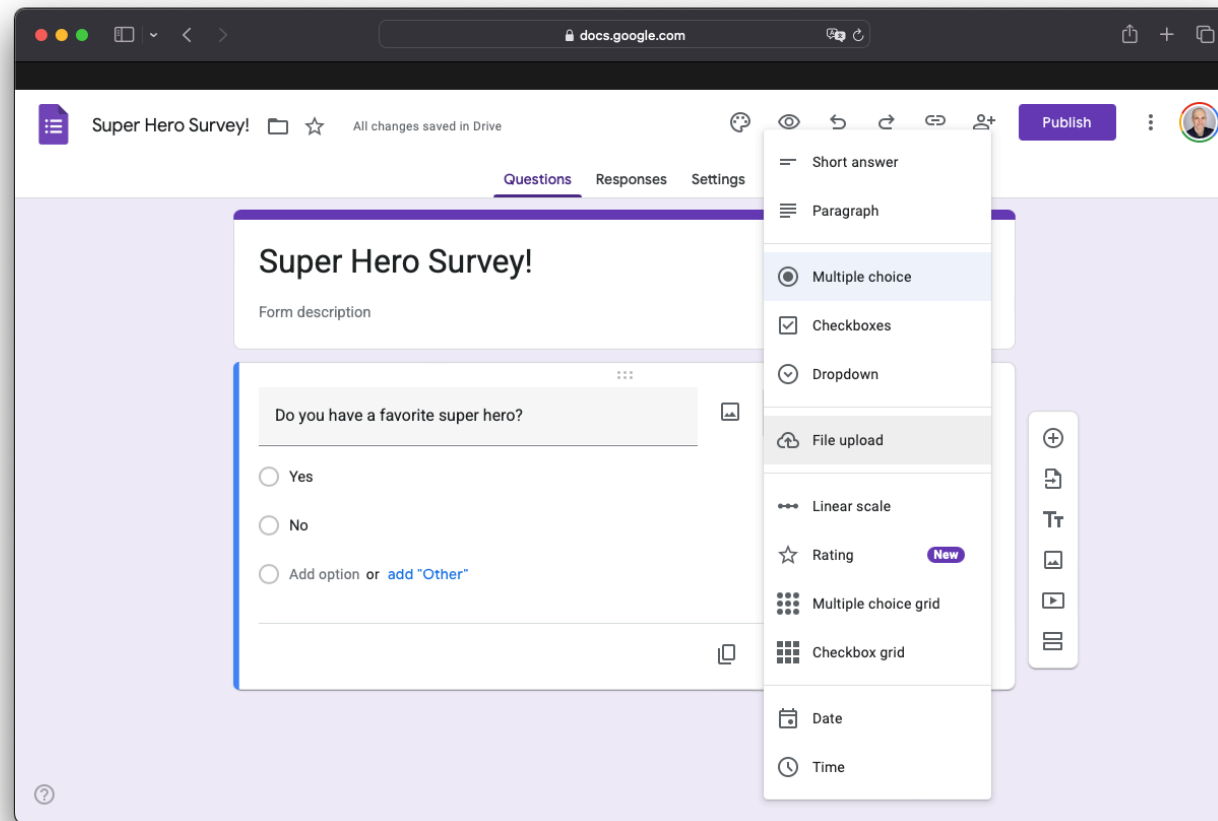
Do you have a favorite super hero? *

☐ Yes

☒ No

Next

WYSIWYG Interface



- ⊗ Reproducibility
- ⊗ Version control
- ⊗ Limited features
- ⊗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



Survey design using Quarto and Shiny



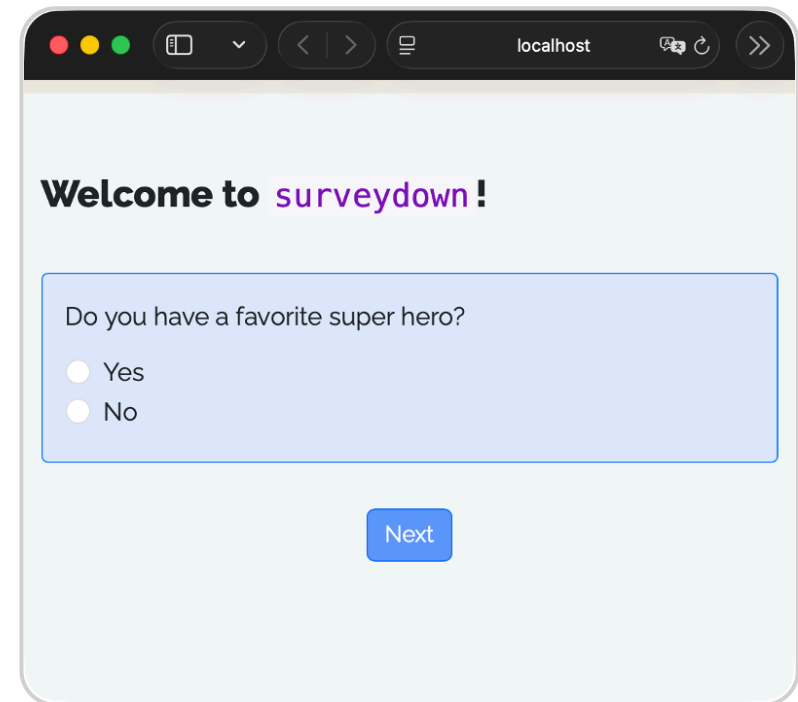
+



+



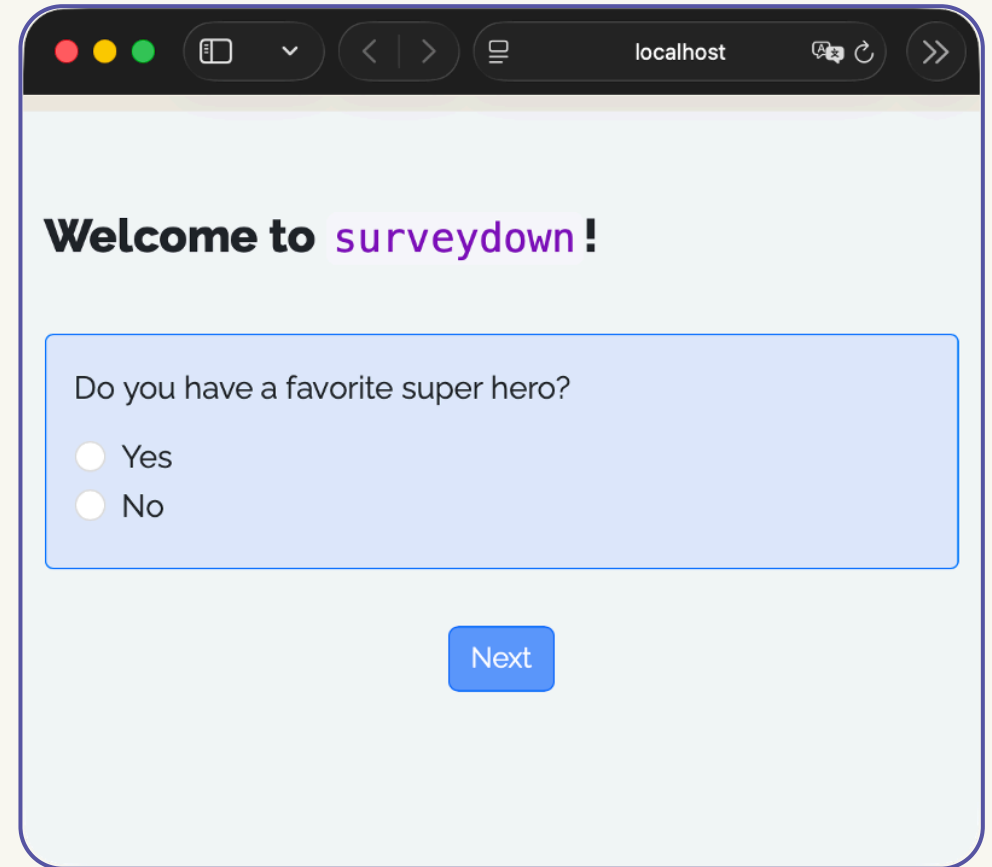
=



survey.qmd

```
1 ---
2 survey-settings:
3   use-cookies: true
4   rate-survey: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id   = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25 ```
26
27 --- some_other_page
28
29 Other content...
```

Rendered Survey



localhost

Welcome to **surveydown**!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

But...



Code-Only Interface

No GUI. Not everyone is comfortable working purely in code.



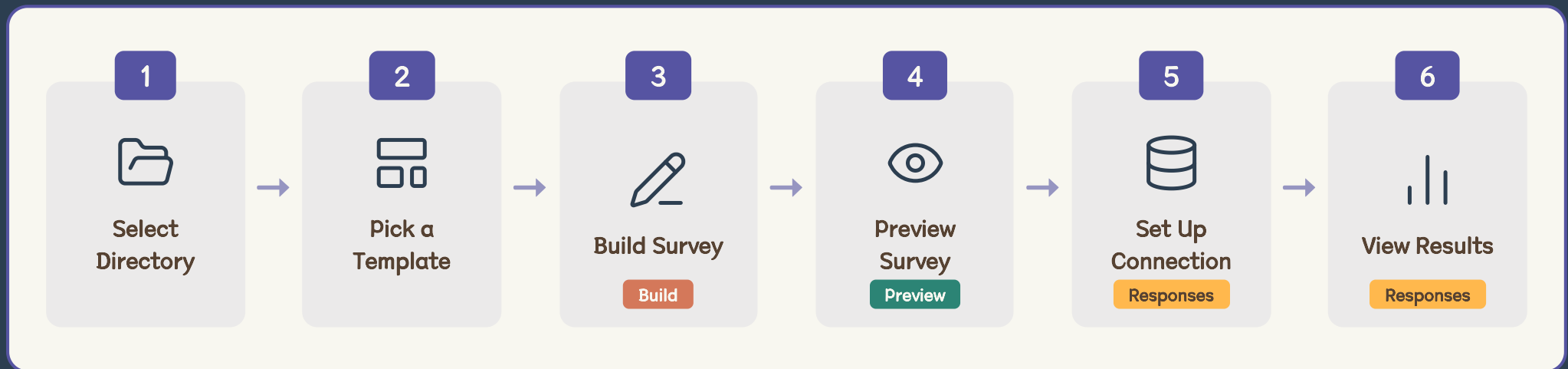
Learning Curve

The survey pipeline is not obvious. It takes time to learn.

Introducing **sdstudio**!



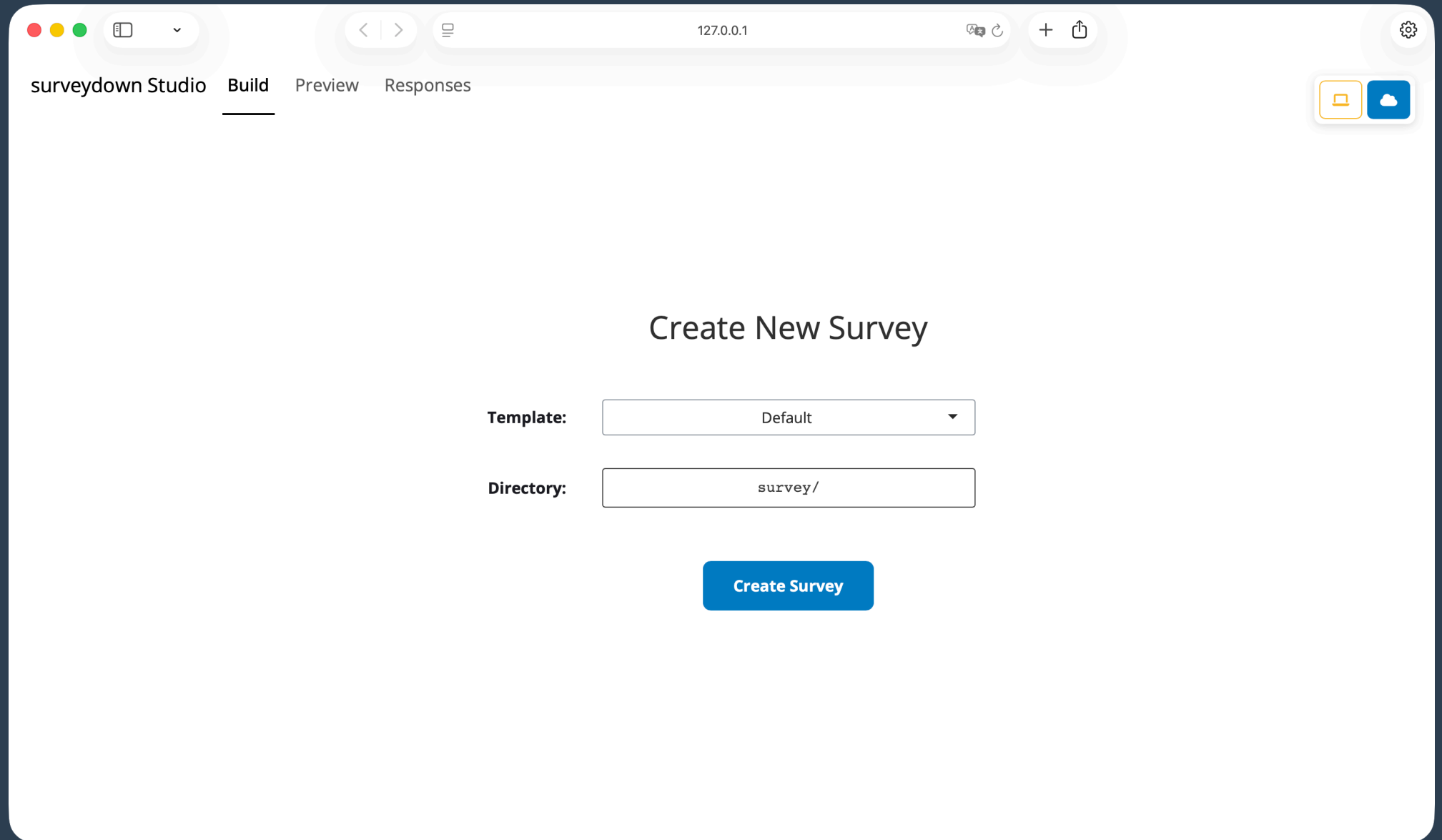
Studio Workflow



Launch with a simple command

```
sdstudio :: launch()
```

Landing Page



The screenshot shows a web application interface for 'surveydown Studio'. The top navigation bar includes 'surveydown Studio', 'Build' (which is underlined), 'Preview', and 'Responses'. To the right of the navigation bar are icons for a folder, a cloud, and a settings gear. The main content area is titled 'Create New Survey'. Below this title are two form fields: 'Template:' with a dropdown menu showing 'Default', and 'Directory:' with a text input field containing 'survey/'. At the bottom of the form is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

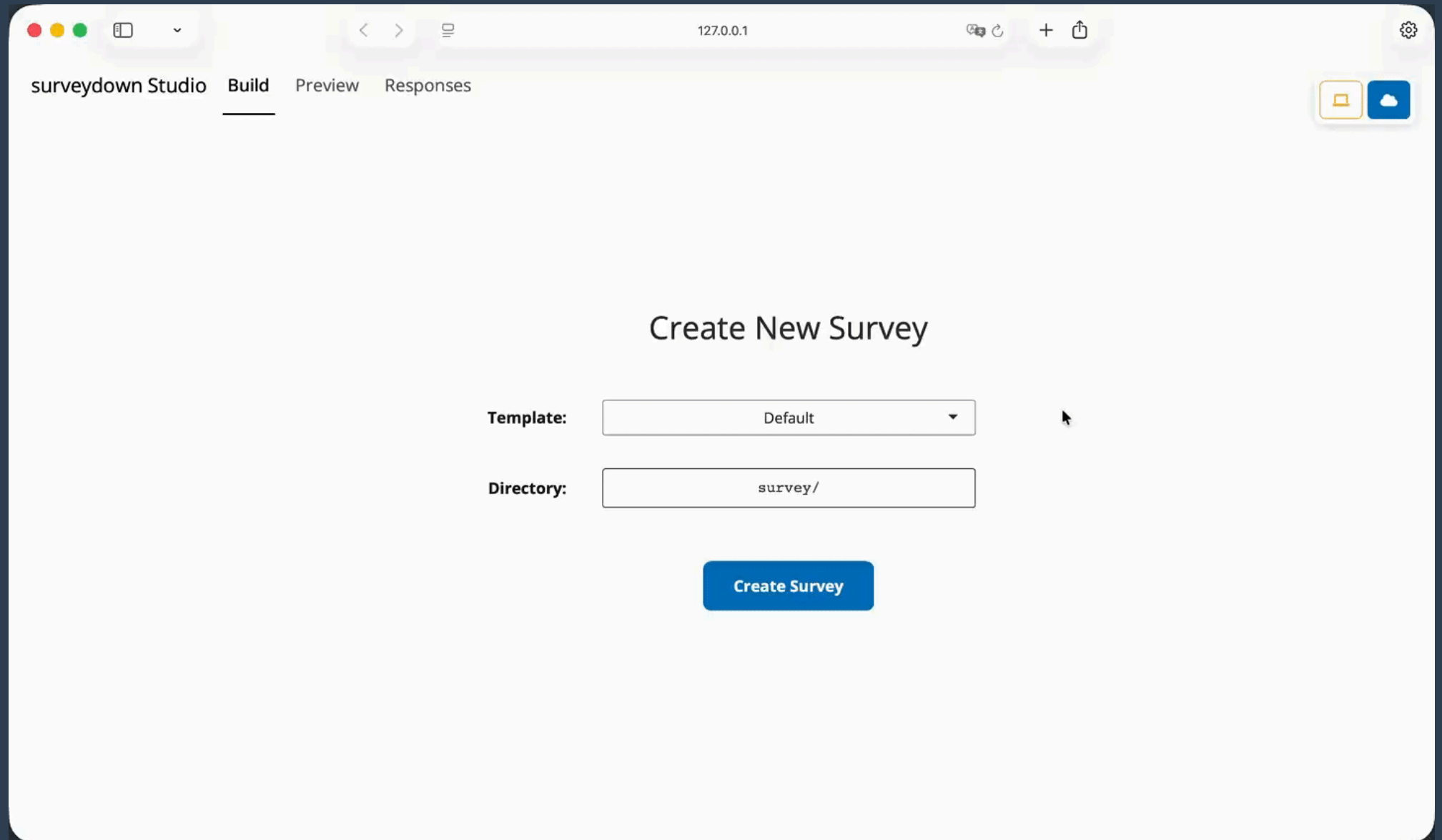
Create New Survey

Template: Default

Directory: survey/

Create Survey

Start with a Template



The screenshot shows the 'surveydown Studio' application window. The title bar includes standard macOS window controls (red, yellow, green buttons) and a dropdown menu. The address bar shows the URL '127.0.0.1'. The application has a top navigation bar with tabs: 'surveydown Studio', 'Build' (which is underlined), 'Preview', and 'Responses'. On the right side of the top bar, there are icons for a folder, a cloud, and a settings gear. The main content area is titled 'Create New Survey'. Below this title, there are two form fields: 'Template:' with a dropdown menu currently showing 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

surveydown.org/templates

surveydown.org

About

Documentation

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FAQ

1 surveydown Studio

Templates – surveydown.org

Templates

Basic

Default Template

Question Types

Questions Defined with YAML

Conditions

Conditional Showing

Conditional Skipping

Conditional Stopping

Randomization

Random Options

Random Options Predefined

Option Shuffling

Reactivity

Reactive Questions

Reactive drilldown

External Redirect

Live Polling

Conjoint

Conjoint Buttons

Conjoint Tables

Custom

Custom Leaflet Map

Custom Plotly Chart

Templates

To help you get started with surveydown, we created some templates that cover many of the features of the package. Feel free to use any as a starting point for your own surveys or as a reference for specific features.

Call `sd_create_survey()` with proper arguments to create your desired templates in your preferred directory. Refer to the [Start with a template](#) section for more details.

Default Template

Welcome to surveydown! This is a default template of a surveydown survey. It has two pages with one question on each page.
Here is a basic "multiple choice" question:

Which type of penguin do you like the best?

Adelie

Chinstrap

Gentoo

You need to insert next buttons with `sd_next()` on each page. By default, it will go to the next page defined in the `survey.yml` file, but you can also specify the page with the `next_page` argument set to the name of the page you want to go to next.

Next

Made with surveydown

Source Code

Default Template

BASIC

A minimum template for starting from scratch.

Template - Question Types

Hi! This is a showcase of different question types and question formatting that surveydown supports. Please make your choice below:

Where do you want to go to?
Take me to the page on Question Types the next page
Take me to the page on Question Formatting
Skip to the end

Next

Made with surveydown

Source Code

Question Types

BASIC

A template showcasing all built-in [question types](#).

Template - Questions Defined with YAML

This is a showcase of defining questions with a `questions.yml` file.
The `sd_question()` allows a `yml` argument default to `"questions.yml"`. If you have such file in your project root folder, you can use it to support your question definition.
You can also change to other file name or directory and define questions like this:

```
sd_question(
  id = "your_question_id",
  yml = "folder_name/files_name.yml"
)
```


Please click on the "Next" button to see the demo questions.

Next

Made with surveydown

Source Code

Questions Defined with YAML

BASIC

A template showcasing all built-in [question types](#) using a YAML file.

Categories

All (17)

Basic (3)

Conditions (3)

Conjoint (2)

Custom (2)

Randomization (3)

Reactivity (4)

Site made with   and  quarto

Choose Your Directory

The screenshot shows a web application interface for 'surveydown Studio'. At the top, there is a navigation bar with tabs for 'Build', 'Preview', and 'Responses', with 'Build' being the active tab. The main content area is titled 'Create New Survey'. It contains two form fields: 'Template:' with a dropdown menu set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

The Build Tab

surveydown Studio Build Preview Responses

Structure

- Page: welcome
 - # Default Template Welcome to...
 - penguins | mc
What's your favorite penguin?
 - There are several ways to...
- Page: page2
 - This is another page in...
 - silly_word | text
Write a silly word here:
 - On this page, we don't...
- Page: end
 - ## End This is the...

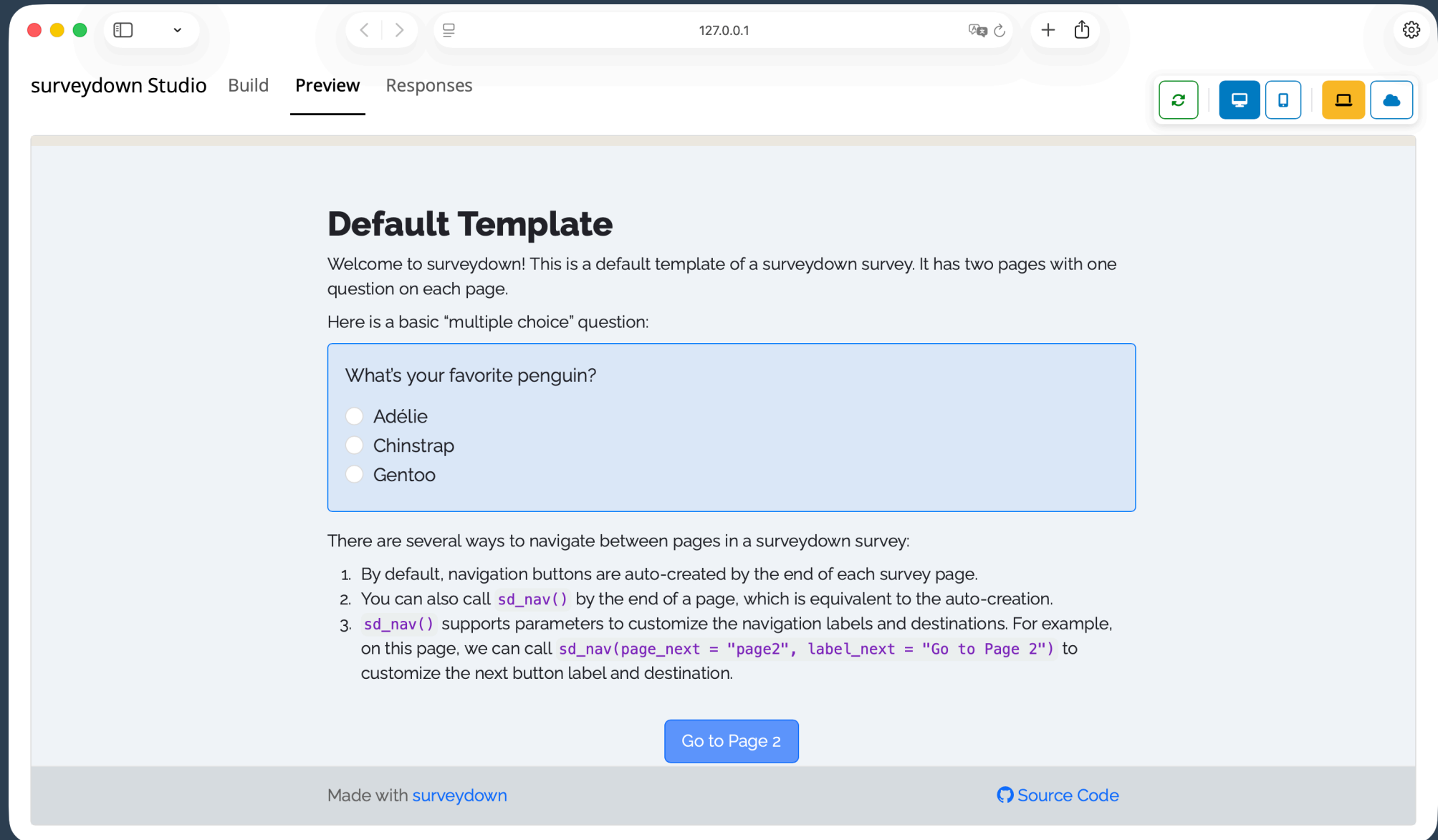
Add A New Page

Code

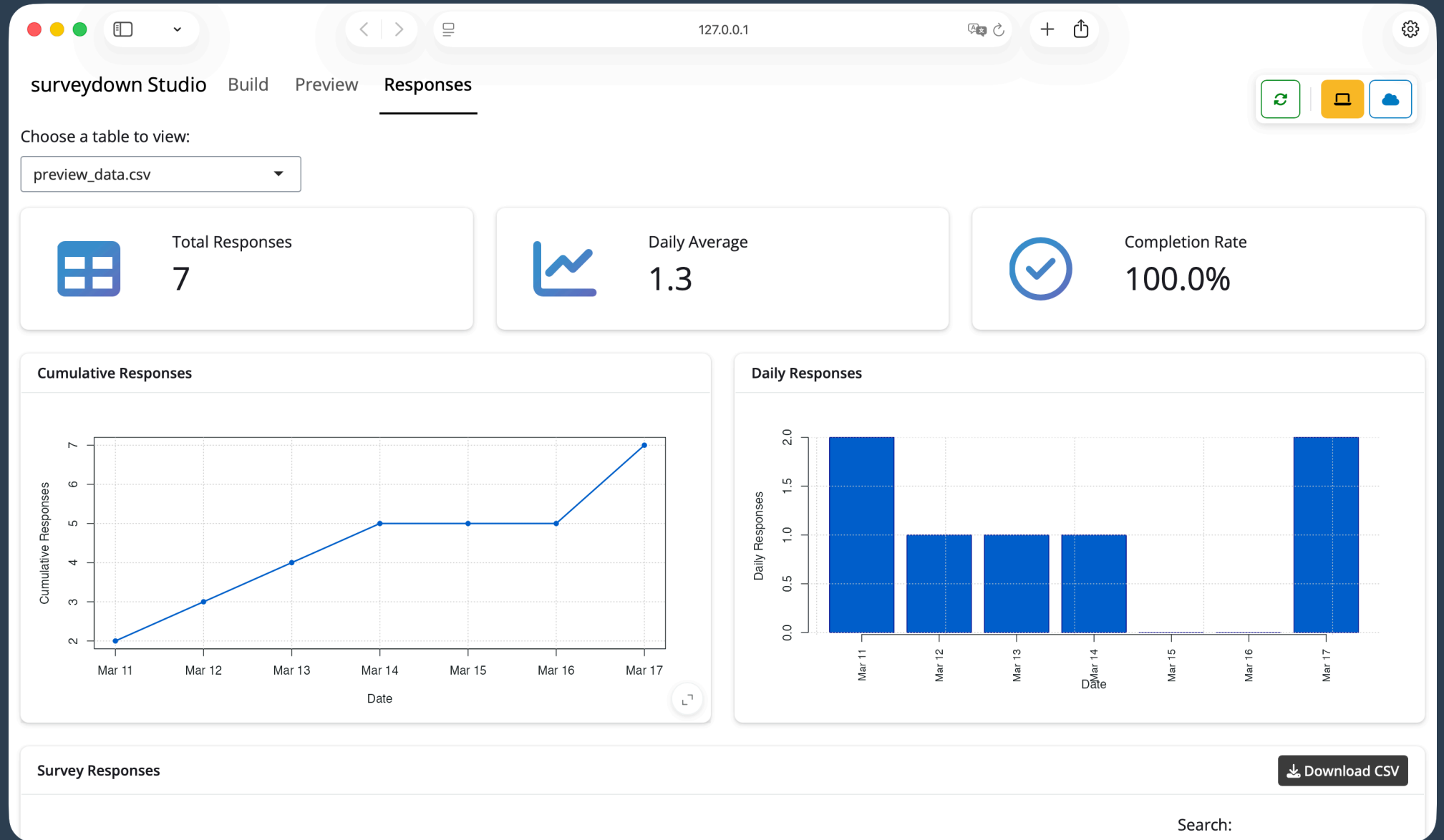
survey.qmd app.R

```
37 --- welcome
38
39 # Default Template
40
41 Welcome to surveydown! This is a default template of a surveydown survey. It has two
42 pages with one question on each page.
43
44 Here is a basic "multiple choice" question:
45
46 ```{r}
47 sd_question(
48   type = 'mc',
49   id   = 'penguins',
50   label = "What's your favorite penguin?",
51   option = c(
52     'Adélie' = 'adelie',
53     'Chinstrap' = 'chinstrap',
54     'Gentoo' = 'gentoo'
55   )
56 )
57
58 There are several ways to navigate between pages in a surveydown survey:
59
60 1. By default, navigation buttons are auto-created by the end of each survey page.
61 2. You can also call `sd_nav()` by the end of a page, which is equivalent to the
62    auto-creation.
63 3. `sd_nav()` supports parameters to customize the navigation labels and destinations.
64    For example, on this page, we can call `sd_nav(page_next = "page2", label_next = "Go to
65    Page 2")` to customize the next button label and destination.
66
67 ```{r}
```

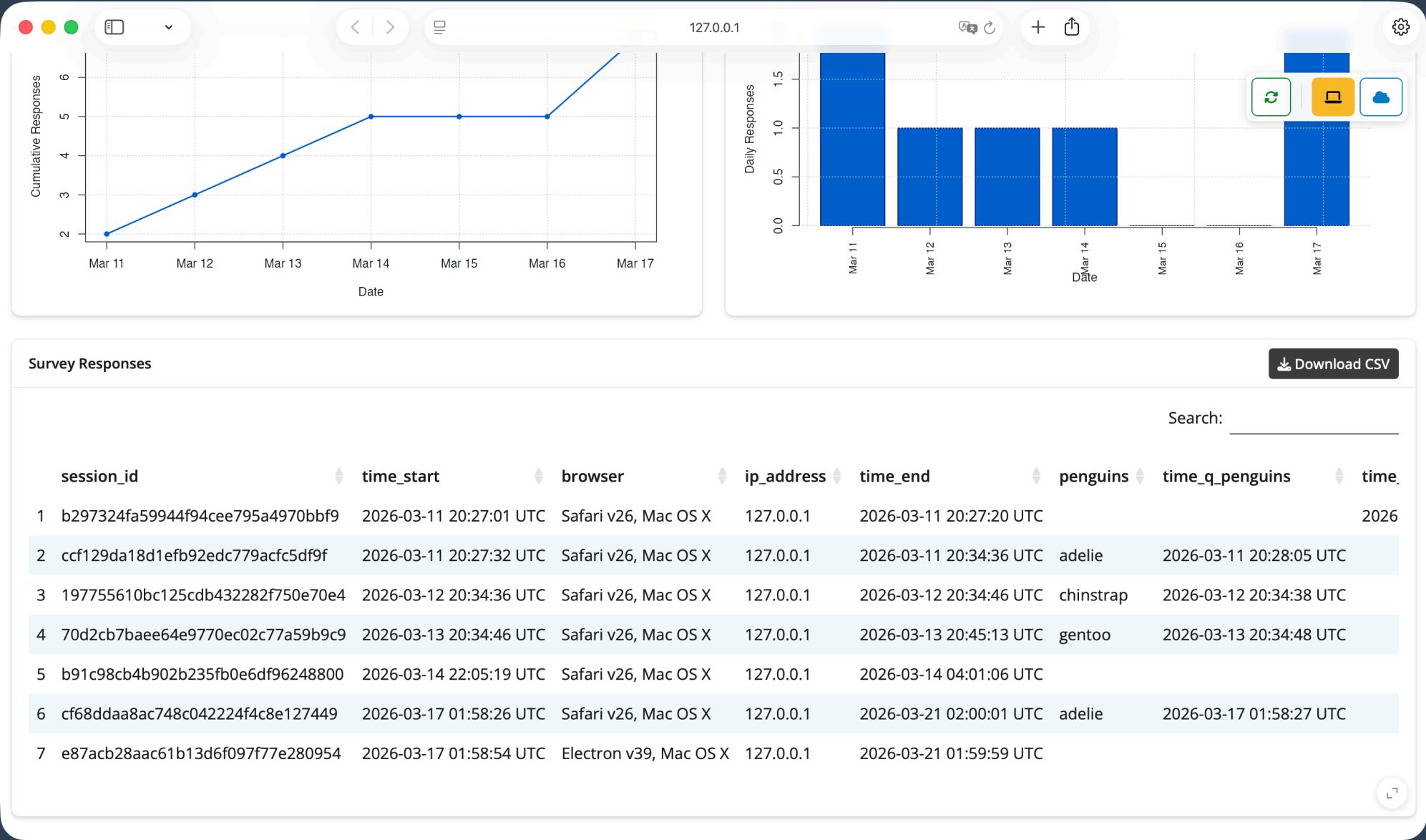

The Preview Tab



The Responses Tab

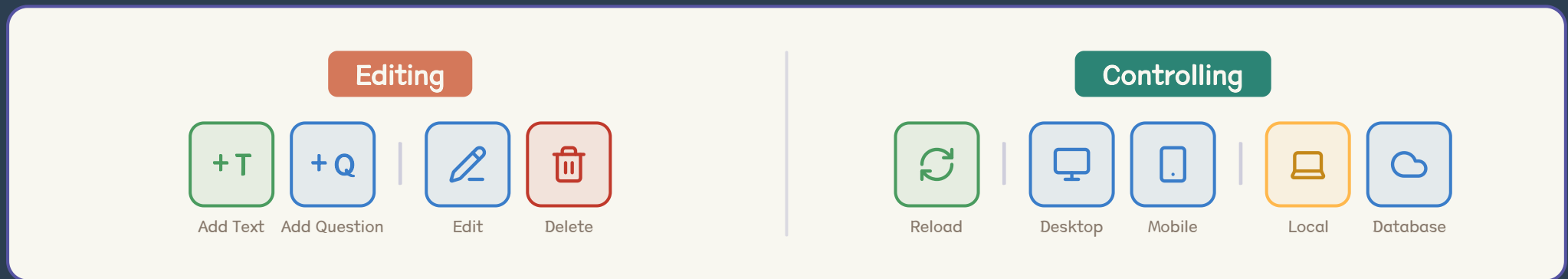


The Responses Tab - Continued



Studio Features

Self-explanatory Icons



Editing

GUI



Point & Click

No coding required

or

Script



Edit Directly

Full code control

GUI - Toggling

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- The "Structure" tab is active, showing a hierarchical view of the survey pages.
- Under "Page: page1", there is a text element with the label "# Title Some texts." and a multiple-choice question element with the label "penguins | mc" and the text "What's your favorite penguin?".
- Under "Page: end", there is a text element with the label "## End This is the...".
- At the bottom of the Structure panel, there is a button labeled "Add A New Page".

Code Panel:

- The "Code" tab is active, showing the R code for the survey.
- The code is for a file named "survey.qmd" and is written in R.
- The code includes the following lines:

```
35 library(surveydown)
34- 
35
36
37 --- page1
38
39- # Title
40
41 Some texts.
42
43- ```{r}
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54- 
55
56 --- end
57
58- ## End
59
60 This is the last page in the survey.
61
62- ```{r}
63 sd_close()
64- ```
```

GUI - Content Editing

The screenshot displays the surveydown Studio interface, which is used for creating and editing surveys. The interface is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title** Some texts.
 - penguins | mc**
What's your favorite penguin?
- Page: end**
 - ## End** This is the...

Code Panel:

```
33 library(surveydown)
34- 
35
36
37 --- page1
38
39- # Title
40
41 Some texts.
42
43- ```{r}
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54- 
55
56 --- end
57
58- ## End
59
60 This is the last page in the survey.
61
62- ```{r}
63 sd_close()
64- 
```


GUI - Content Addition

The screenshot displays the surveydown Studio interface, which is used for creating interactive surveys. The interface is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title** New texts.
 - penguins | mc**
What's your favorite penguin?
- Page: end**
 - ## End** This is the...

An orange button labeled "Add A New Page" is located at the bottom of the Structure panel.

Code Panel:

The Code panel shows the R code for the survey. The code is as follows:

```
33 library(surveydown)
34 
35 
36 
37 --- page1
38 
39 # Title
40 
41 New texts.
42 
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 
55 
56 --- end
57 
58 ## End
59 
60 This is the last page in the survey.
61 
62 ```{r}
63 sd_close()
64 
```

GUI - Page Addition

The screenshot displays the surveydown Studio interface. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Structure' panel on the left shows a hierarchical view of the survey pages. It lists 'Page: page1' with a title and two questions (multiple choice and text), followed by 'Page: end' with a closing text. An 'Add A New Page' button is at the bottom. The 'Code' panel on the right shows the corresponding R code for the survey, including question definitions and page transitions.

Structure Panel:

- Page: page1
 - # Title New texts.
 - penguins | mc
What's your favorite penguin?
 - text_question | text
Write something:
- Page: end
 - ## End This is the...

Code Panel:

```
survey.qmd  app.R
41 new texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ```
55 ```{r}
56 sd_question(
57   type = "text",
58   id   = "text_question",
59   label = "Write something:"
60 )
61 ```
62
63
64 --- end
65
66 ## End
67
68 This is the last page in the survey.
69
70 ```{r}
71 sd_close()
72 ```
```

GUI - Page Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel (Left):

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- Below the tabs is a "Structure" header with a refresh icon and a share icon.
- The structure is organized into three pages:
 - Page: page1**: Contains three items:
 - A title block: "# Title New texts."
 - A multiple-choice question: "penguins | mc" with the label "What's your favorite penguin?".
 - A text question: "text_question | text" with the label "Write something:".
 - Page: page2**: Contains one item:
 - A text block: "This is page 2."
 - Page: end**: Contains one item:
 - A text block: "## End This is the..."
- At the bottom of the structure panel is a button labeled "Add A New Page".

Code Panel (Right):

- The header shows "Code" and two tabs: "survey.qmd" and "app.R".
- The "survey.qmd" tab is active, showing the following code:

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61 ```  
62  
63  
64 --- page2  
65  
66 This is page 2.  
67  
68  
69 --- end  
70  
71 ## End  
72
```

GUI - Content Relocation

The screenshot displays the surveydown Studio interface, which is used for creating and managing surveys. The interface is divided into two main panels: Structure and Code.

Structure Panel: This panel shows the hierarchical structure of the survey. It includes three pages: "Page: page1", "Page: page2", and "Page: end". Each page contains a list of questions and their types. For example, "Page: page1" has a title question, a multiple-choice question about favorite penguins, and a text question. "Page: page2" has a single text question. The "Page: end" section contains a final text question. A button at the bottom of the Structure panel allows adding a new page.

Code Panel: This panel shows the R code that generates the survey. The code is organized into sections corresponding to the pages in the Structure panel. It includes functions like `sd_question` to define questions and their options. The code is written in R and uses the `survey` package.

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id = "text_question",  
59   label = "Write something:"  
60 )  
61 ```  
62  
63  
64 --- page2  
65  
66 This is page 2.  
67  
68  
69 --- end  
70  
71 ## End  
72
```

Script Editing

The screenshot displays the surveydown Studio web application. The interface is divided into two main panels: 'Structure' on the left and 'Code' on the right. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses' tabs, with 'Build' currently selected. The 'Structure' panel shows a hierarchical view of the survey pages. It includes three pages: 'Page: page1', 'Page: page2', and 'Page: end'. Each page contains a list of questions or text blocks with icons for editing and deleting. For example, 'Page: page1' has a title 'New texts.' and a multiple-choice question 'What's your favorite penguin?'. 'Page: page2' has a text block 'This is page 2.' and a text question 'Write something:'. The 'Page: end' block is partially visible. An orange button at the bottom of the Structure panel says 'Add A New Page'. The 'Code' panel shows the R script for the survey. It includes a function 'sd_question' that defines the survey structure. The script is divided into sections for 'page1', 'page2', and 'end'. The 'page1' section defines a multiple-choice question about penguins. The 'page2' section defines a text question. The 'end' section defines the final text block. The script is currently editing the 'page2' section.

surveydown Studio Build Preview Responses

Structure

Page: page1

- # Title New texts.
- penguins | mc
What's your favorite penguin?

Page: page2

- This is page 2.
- text_question | text
Write something:

Page: end

- ## End This is the...

Add A New Page

Code

```
survey.qmd app.R
44 sd_question(
45   type = 'mc',
46   id   = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55 --- page2
56
57 This is page 2.
58
59 {r}
60 sd_question(
61   type = "text",
62   id   = "text_question",
63   label = "Write something:"
64 )
65 ---
66
67 --- end
68
69 ## End
70
71 This is the last page in the survey.
72
73 {r}
74 sd_close()
75
```


Instant Preview

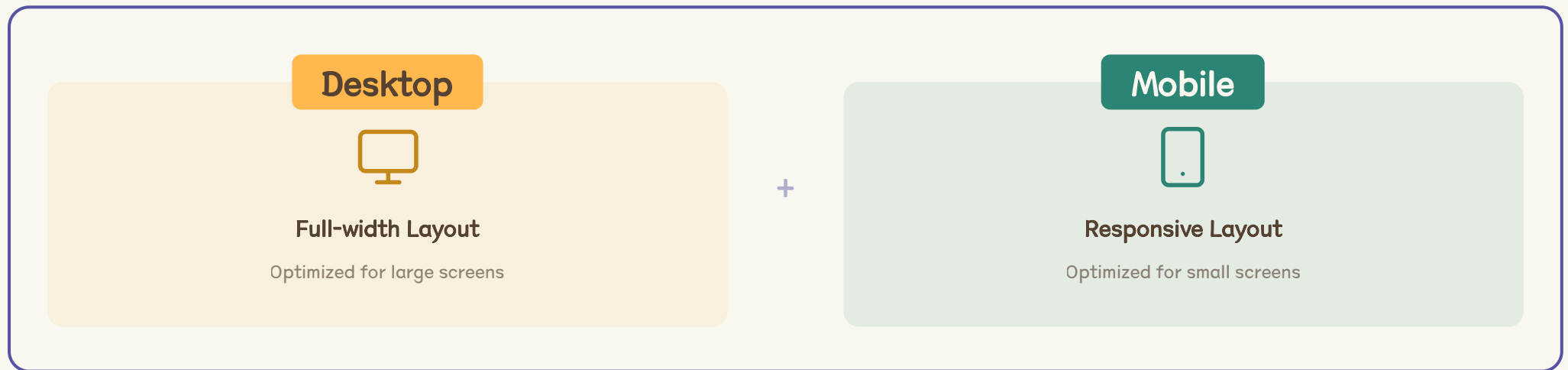
The screenshot displays the surveydown Studio application interface. At the top, there's a navigation bar with tabs for 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Build' tab is currently active. Below the navigation bar, the interface is split into two main panels: 'Structure' on the left and 'Code' on the right.

Structure Panel: This panel shows a hierarchical view of the survey structure. It lists three pages: 'Page: page1', 'Page: page2', and 'Page: end'. Each page has associated elements: 'Page: page1' contains a title '# Title New texts.' and a multiple-choice question 'penguins | mc' with the label 'What's your favorite penguin?'. 'Page: page2' contains a text element 'This is page 2.' and a text question 'text_question | text' with the label 'Write something:'. 'Page: end' contains a closing text element '## End This is the...'. Each element has icons for adding, editing, and deleting.

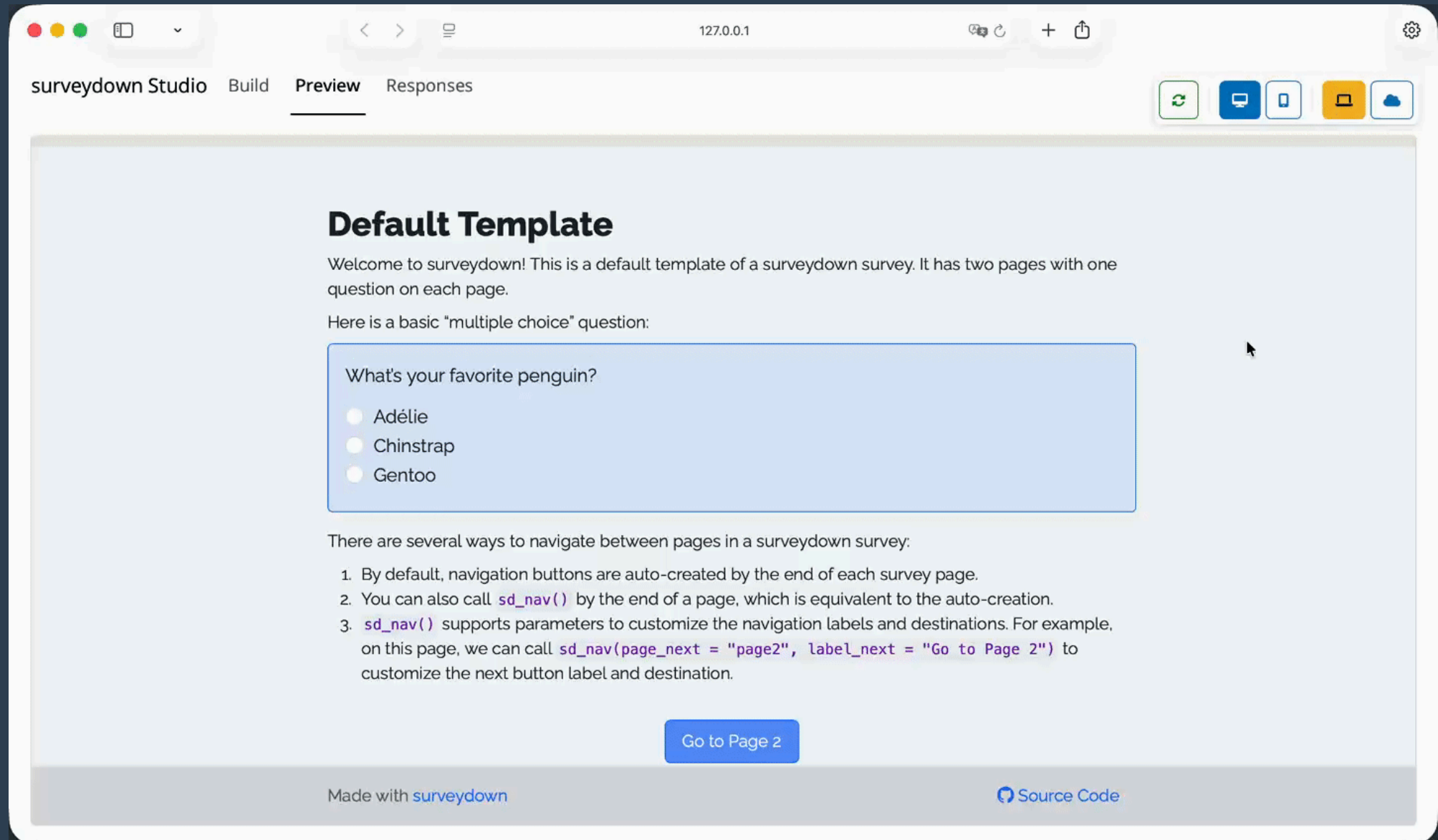
Code Panel: This panel shows the R code for the survey. It includes a file explorer at the top with 'survey.qmd' and 'app.R'. The code is written in R and defines the survey structure using the 'sd_question' function. It includes comments for the title and questions, and defines the options for the multiple-choice question.

```
34 -  
35 -  
36 -  
37 --- page1  
38 -  
39 # Title  
40 -  
41 New texts.  
42 -  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 -  
55 --- page2  
56 -  
57 This is page 2.  
58 -  
59 ```{r}  
60 sd_question(  
61   type = "text",  
62   id = "text_question",  
63   label = "Write something:"  
64 )  
65 -
```

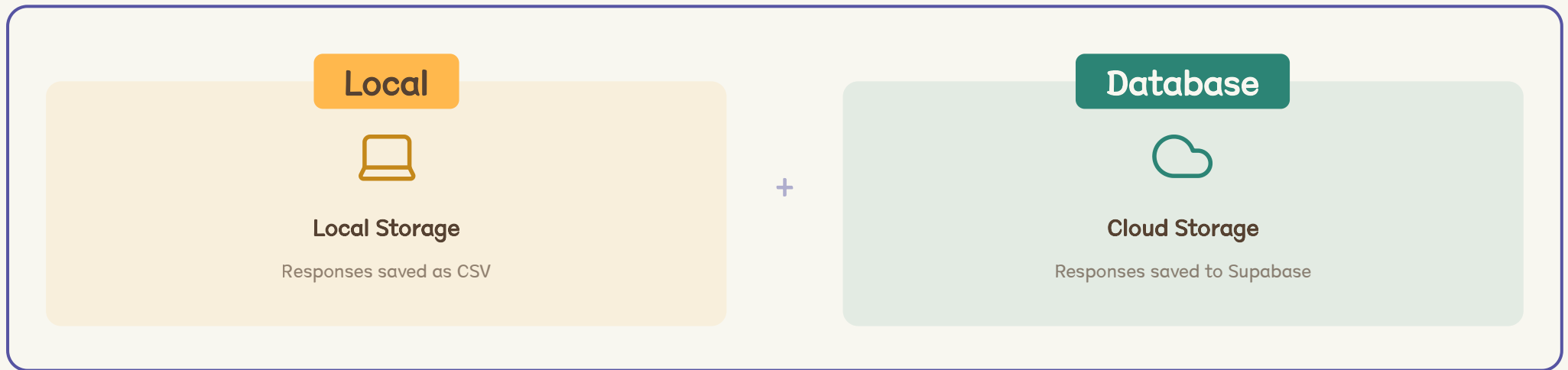
Views



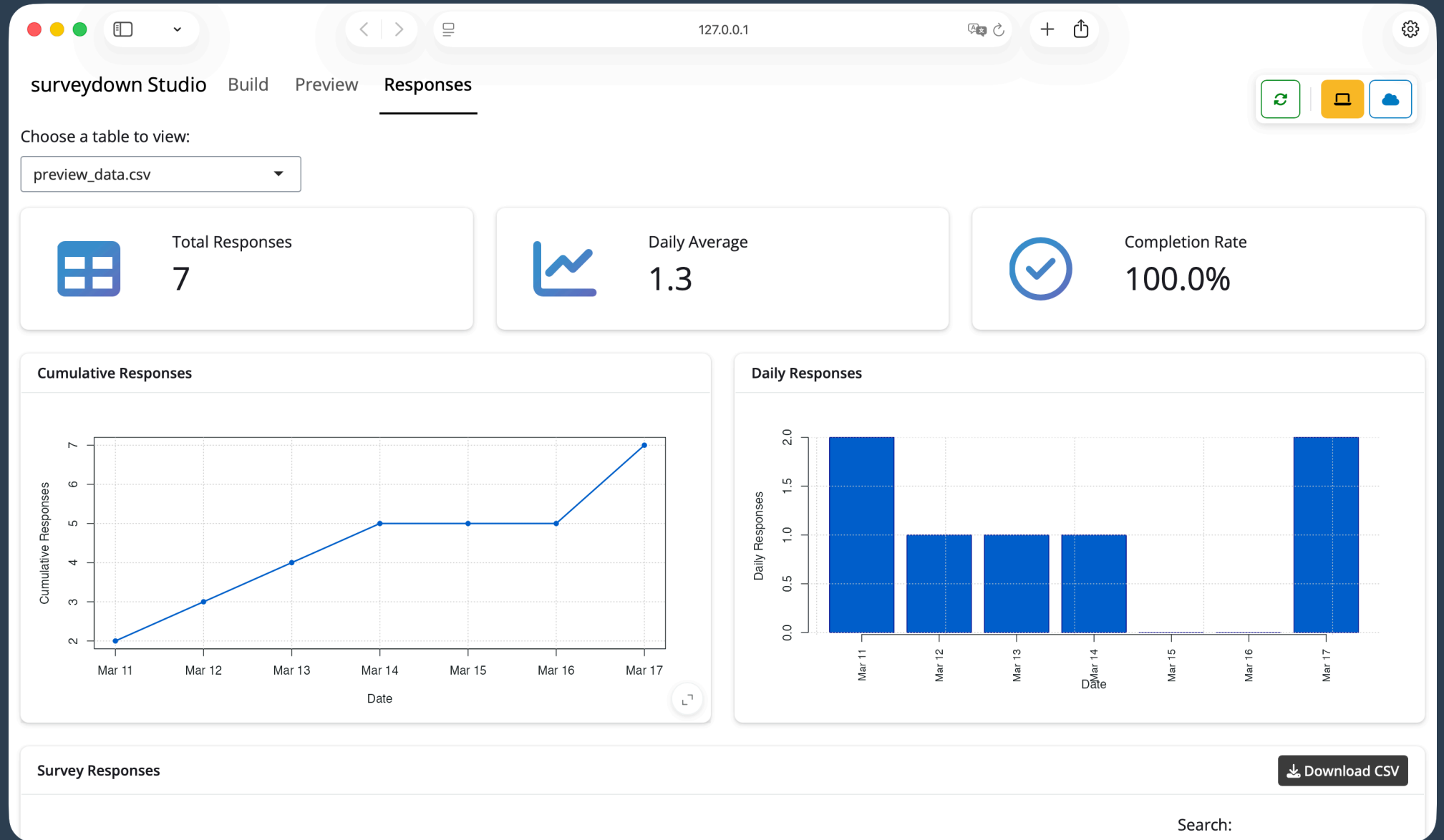
Desktop and Mobile Views



Responses



Local Responses



Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Close

Not connected

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

default

Test Connection

Choose a table to view:

Select or type table name...

Database Responses

surveydown Studio

Build

Preview

Responses

127.0.0.1

Connected (GSSAPI disabled)

Database Connection

Close

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

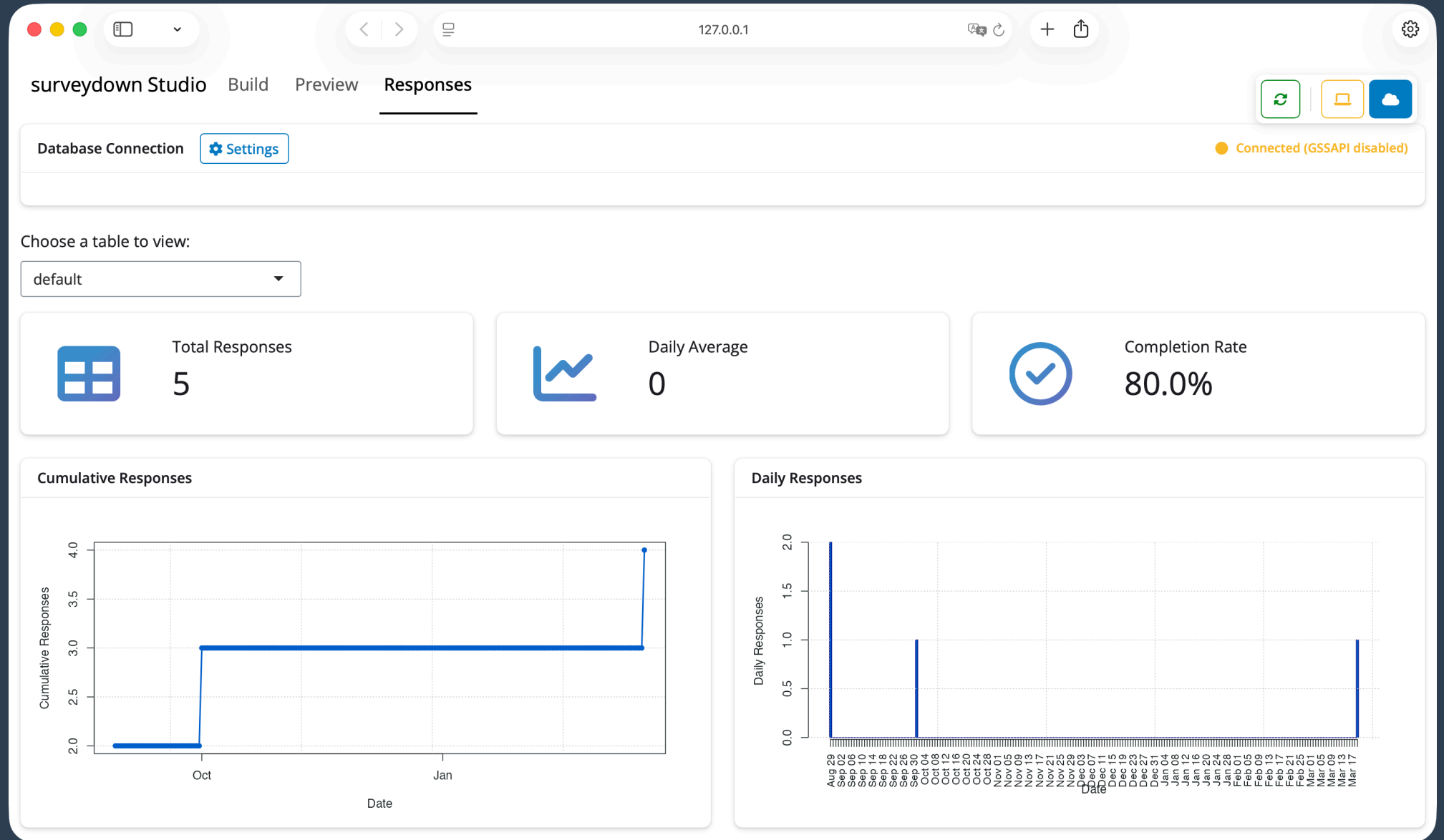
default

Test Connection

Choose a table to view:

default

Database Responses



sdstudio gives you...



GUI for Survey Editing



Live Preview



Response Dashboard



SQL Connection



Data View & Download



...and more!

surveydown + sdstudio

surveydown

 Text-based editing

 Published paper

 8000+ CRAN downloads

sdstudio

 GUI editing & live preview

 Actively developed

Open to contributions

 Better together!

Ways to help:

github.com/surveydown-dev/sdstudio/

- Try it out!
- Give it a ★
- Post an issue
- Join the GitHub discussion
- Contribute new features



Meet the team!

surveydown.org



John Helveston, Ph.D.



Pingfan Hu



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